

1 List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Motherboard - The Main Circuit Board. It connects and communicates between RAM, GPU, CPU and other components.

CPU – The Central Processing Unit, also called the Processor. It performs calculations and executes instructions to run a program and the OS.

GPU -The Graphical Processing Unit handles graphics and video processing. It can be integrated into the CPU or be a dedicated graphics card.

RAM – It is Random Access Memory. It's temporary memory. It can be 4GB, 8GB, 16GB, 32GB.

HDD – It is a hard disk where all the data is saved of files, application software, OS. It can be 500GB up to 22TB.

CMOS Cell – A small battery on motherboard which stores date and time and also provides power supply to components.

CPU Fan – A cooling fan mounted near the CPU. It removes the processor and helps to prevent overheating.

Network card – Allows computer to connect to a network or the internet or the internet using Ethernet or WiFi.

2 Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

### Step 1: Shutdown and Disconnect the PC

Turn off the computer and remove the plug from a cabinet and also remove VGA cable from monitor and also remove keyboard, CPU.

### Step 2 Open The Cabinet



Remove screws holding the side panel and carefully slide panel off.

### Step 3: Remove RAM



Push down the clips on both sides of the RAM slot and carefully remove the RAM module by holding it from the edges.

#### **Step 4: Remove Storage Devices**



Disconnect the cables and remove the HDD/SSD from its mounting bracket or slot.

#### **Step 5: Remove the CPU Cooler**



Disconnect the CPU fan cable from the motherboard and carefully remove the CPU cooler/heatsink according to its mounting mechanism.

#### **Step 6: Remove the CPU**



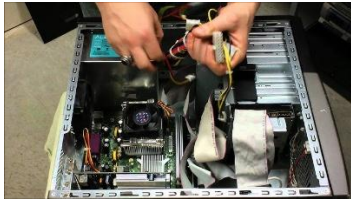
Release the CPU socket locking mechanism and carefully lift the CPU out without touching its pins or contacts.

#### **Step 7: Remove the CMOS Battery**



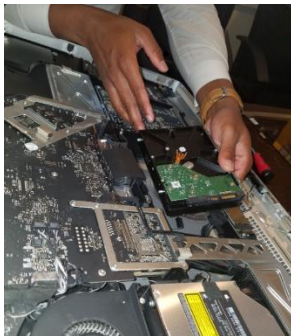
Carefully release the clip holding the CMOS battery and remove the battery from the motherboard.

### **Step 8: Remove the Power Supply**



Disconnect all PSU cables from the components. Remove the screws securing the power supply to the cabinet and carefully take it out.

### **Step 9: Remove the Motherboard**



Disconnect any remaining cables and remove the screws securing the motherboard to the cabinet. Carefully lift the motherboard out.

3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

### **Step 1: Install the Motherboard**

Place the motherboard carefully inside the cabinet and secure it with the appropriate screws. The motherboard is installed first because most other components connect directly to it.

### **Step 2: Install the CPU**

Place the CPU correctly into the CPU socket and lock the socket mechanism. The CPU must be installed before the CPU cooler because the cooler sits directly on top of it.

### **Step 3: Install the CPU Cooler/Fan**

Attach the CPU cooler to the CPU and connect its fan cable to the CPU\_FAN header on the motherboard. This prevents the processor from overheating during operation.

### **Step 4: Install the RAM**

Insert the RAM modules into the correct motherboard slots until the clips lock into place. RAM must be properly installed before testing or starting the PC.

### **Step 5: Install the GPU**

Insert the graphics card into the appropriate PCIe slot and secure it to the cabinet with screws. Connect its required power cable from the PSU.

### **Step 6: Install the HDD/SSD**

Place the storage drive in its mounting position and secure it. Connect the required SATA/data cable to the motherboard and power cable from the PSU.

**Step 7: Install the Power Supply (PSU)**

Place the PSU in the cabinet and secure it with screws. Connect the motherboard, CPU, GPU, storage, and other required power cables.

**Step 8: Connect Front-Panel and Other Cables**

Connect the cabinet's power switch, reset switch, USB, audio, and other front-panel cables to their appropriate motherboard headers.

**Step 9: Check All Connections**

Make sure RAM, GPU, storage cables, power cables, and fan connectors are properly connected. This step is important to prevent loose connections and hardware problems.

**Step 10: Close the Cabinet**

After checking everything, carefully replace and secure the side panel of the cabinet.

**Step 11: Reconnect External Cables**

Connect the monitor, keyboard, mouse, Ethernet cable, and power cable to the PC.

**Step 12: Test the PC**

Turn on the PC and check whether it starts normally. Verify that the CPU fan, GPU, display, storage, and other components are working correctly.

**Why Orders Matters**

**The motherboard, CPU, cooler, RAM, GPU, and storage should be installed in a logical sequence because many components connect directly to the motherboard, and some parts can block access to others if installed too early. Checking all connections before powering on also helps prevent damage and troubleshooting problems.**

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

**Tools Needed**

1. **Phillips-head screwdriver** – Used to remove and tighten screws inside the PC cabinet.
2. **Flat-head screwdriver** – Used for certain screws and clips when required.
3. **Anti-static wrist strap** – Protects PC components from static electricity.
4. **Small container** – Keeps screws and small parts safely together.
5. **Compressed air** – Used to remove dust from fans, heatsinks, and other components.
6. **Cable ties** – Used to organize cables and improve airflow.

**Safety Precautions**

1. **Turn off the PC** – Prevents electrical and hardware damage.
2. **Unplug the power cable** – Reduces the risk of electric shock.
3. **Disconnect all external cables** – Prevents damage to connected devices and ports.

4. **Work on a clean and dry surface** – Keeps the components safe from dust and moisture.
5. **Use an anti-static wrist strap** – Prevents static electricity from damaging sensitive components.
6. **Hold components by their edges** – Protects the contacts and circuits from damage.
7. **Do not force any component** – Prevents damage to slots, connectors, and the motherboard.
8. **Keep screws safely** – Prevents screws from being lost or misplaced.
9. **Handle the CPU carefully** – Prevents damage to the CPU pins or contacts.
10. **Check all connections before switching on** – Ensures that all components are installed correctly and safely.